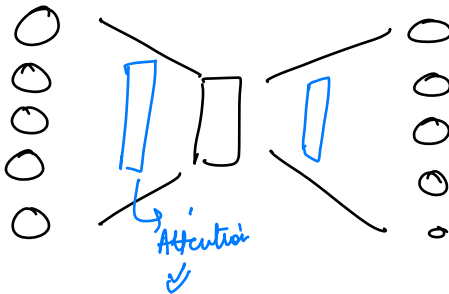


Agenda → Attention
+ self attention
+ Multi-head attention

{ Transformers → Best → GPT, Claude,
Llama, {
⋮

Amazon Alexa
↳ Language translation

LSTM/RNN Based encoder-decoder
↳ Machine translation.



→ RNN

→ Sequential bottleneck

→ long range dependencies could be tricky



Training would be slow

TRANSFORMERS

→ Would remove the recurrence

→ Self attention

✓ → Positional encoding (Small GPS coordinates)

Self attention / Attention



dog bites man or man bites dog

RNN h_{t-1} h_t ...

Transformers Compute Self Attention / Attention for all tokens in parallel

"I love Machine Learning"



Tokenisation ["I", "love", "Machine", "Learning"]



Embedding (word 2 vec, ...) Embedding vector
[512 dim]

$I \rightarrow e_1$

$love \rightarrow e_2$



Positional Embeddings

$I \rightarrow e_1 + p_1 \rightarrow \text{at pos 0}$

$love \rightarrow e_2 + p_2 \rightarrow \text{at pos 1}$

$Machine \rightarrow e_3 + p_3 \rightarrow \text{at pos 2}$

⋮

⋮

(Sine | Cos formula)

Attention → Self Attention

- ① Compare Q with all keys in the sentence.
- ② Scores = Dot product of Q & K
- ③ Softmax over Scores → attention weights (a)
- ④ Content vector → $Q \times \underline{V}$

Effect

"Learning" attends strongly to "Machine"

"I" attend to both "ML & love"



Multi head attention

Head 1 $\rightarrow$ Syntax / Grammar

Head 2 $\rightarrow$ Semantic Similarity

Head 3 $\rightarrow$ next token, previous token / long term dependencies

[Head 1 || Head 2 || Head 3]

Output

$y \rightarrow$ J'adore l'apprentissage automatique

Decoder

Step 1, $\langle s \rangle^{L_0}$

Step 2 $\langle s \rangle J'$

Step 3 $\langle s \rangle J'$ adore

Step 4 $\langle s \rangle J'$ adore I' apprentissage

At each step $\rightarrow$ Allow each position to attend only to position $\leq$ its own index

$\nrightarrow$ Q, K, V is still there, but with a mask that zeros out the future positions.

Why?

no peeking at future words

Cross attention

French word 'aime' attends to
encoder's english representation
(I love machine learning)

English: "I love machine learning"

French: "J'adore l'apprentissage automatique"

Encoder pass

Tokenize English $\rightarrow$ [I, love, machine, learning].

Embeddings: [e1, e2, e3, e4].

PE [P₁, P₂, P₃, P₄]

output $e_1 + P_1, e_2 + P_2, \dots$

Multi-head self attention $\rightarrow$ [understanding]

Feed forward layer

Decoder pass

$\langle s \rangle$

Embeddings + PE

Masked self attention $\rightarrow \langle s \rangle$

Cross attention



Key



Feedforward

linear + Softmax $\rightarrow$ vocab $\rightarrow$ (Je)

Deorden